

Department of Aeronautics

Group Design Project Information for Students and Supervisors: 2024-25

**MEng Aeronautical Engineering (H401)
MEng Aeronautics with Spacecraft Engineering (H415)**

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1. Introduction

The third year large Group Design Project is a significant part of the H401 & H415 programmes, worth 25% of the third year's marks and occupying five full-time weeks immediately following the end of the third year summer exams.

It provides students with the opportunity to learn how to function and contribute creatively in a large multi-disciplinary team, simulating the industrial approach to engineering design and teamwork. With five different challenging projects on offer, students must develop innovative solutions, expanding on their prior knowledge and researching new ideas, while appreciating the constraints imposed in a multi-disciplinary design environment.

2. Contacts

Administrative:

Aeronautics Undergraduate Office: ae.office@imperial.ac.uk

Academic Coordinator:

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3. Role Selection and Allocation

3.1 The 2024-2025 Group Design Projects

Neptune Exploration Mission (NEM25)

Neptune, the furthest confirmed major planet in the Solar System, offers an invaluable opportunity for scientific discovery. Due to its vast distance, Neptune has only been visited once during a flyby by the Voyager 2 spacecraft. Neptune is a planet shrouded in mystery. Winds in its atmosphere are supersonic and reach speeds of more than 2,200 km/h. It is heated internally due to mechanisms that are not well understood, and its magnetic field is curiously tilted. Neptune's largest moon, Triton, is one of the most geologically active Solar System bodies and it might have a subsurface ocean (maybe harboring life?).

The aim of this project is to design the first robotic space exploration mission to explore the Neptunian system and the Kuiper belt. A key element of the mission is a probe that will be deployed from the main spacecraft to descend through Neptune's atmosphere and investigate Neptune's core to gather data. The project team must address all aspects of space mission design, including project management, interaction with key stakeholders, definition and exploration of mission concepts and architectures, and spacecraft and individual subsystem design. This will involve overcoming considerable challenges associated with reaching the furthest planet in the Solar System, making sure that the mission operates reliably, and staying within the budget (sometimes more an art than a science).

Academic Advisors: Drs Amato & Bruce (coordinators), Cantwell, Lubbock, and Profs Aliabadi, Doorly, Laizet, Lee and Montomoli.

Roles Available:

- NEM01 – Project coordination & budget (x4)
- NEM02 – Neptune descent probe subsystem design (x8)
- NEM03 – Neptune orbiter subsystem design (x8)
- NEM04 – Trajectory design (x5)
- NEM05 – Launch & Interplanetary cruise systems (x5)

H2-STOL (HST25)

By the end of the project your team will have designed a 9 passenger hydrogen fuel-cell-powered short take-off and landing aircraft. The aircraft will be designed to operate for transportation and tourist operations in Canada, Alaska, New Zealand and remote regions of South East Asia. It should have a take-off and landing distance of around 250 m and a range of 700 nmi.

Oh, and we would also like it to be able to operate from lakes and other still bodies of water.

Academic Advisors: Dr Li, Liem, Panesar and Professors Chernyshenko, Hewson (coordinator), Mainini, Morrison and Palacios

Roles Available:

- HST01: Configuration Definition (x3)
- HST02: Propulsion System (x3)
- HST03: Aerodynamic Design (x2)
- HST04: Structural Design and Manufacturing (x3)
- HST05: Flight Dynamics and Control (x3)
- HST06: Safety, Systems and Ground Infrastructure (x3)
- HST07: Hydrodynamics (x3)

Shell Eco Marathon (SEM25)

The aim of the GDP is to design a 3-wheel prototype ultra-efficient, lightweight battery electric vehicle to compete in Shell Eco-marathon Europe and Africa. The competition takes place in the Silesia ring track in Poland. The objective is to complete 11 laps of the track (15.620 km) at an average speed above 25 km/h, i.e. in a time within 39 minutes.

The design will be a tadpole configuration (2 front wheels for steering and 1 rear wheel for power), which will use lithium-ion batteries and upgrade an existing chassis. The main purpose of the upgrade is to improve the aerodynamic performance by reducing flow separation, and to incorporate a small ventilator for driver comfort and, potentially, flow control. The final design should cost £20—25K (Imperial Racing Green Team allocated SEM budget).

The rules of the competition are available at: <https://www.shellecomarathon.com/about/global-rules.html>

Academic Advisors: Drs Billoti, MacCarthy, Tagarielli, Venetsanos, and Profs Papadakis, Peiró (coordinator).

Roles Available:

- SEM01 – Technical director: project coordination, race strategy and rules compliance (x1)
- SEM02 – Packaging, Integration and CAD (x3)
- SEM03 – Aeroshell aerodynamics (x4)
- SEM04 – Aeroshell structural analysis and manufacturing (x4)
- SEM05 – Powertrain (x2)
- SEM06 – Lap simulation, vehicle dynamics and traction (x2)
- SEM07 – Dashboard, communications, and telemetry (x1)
- SEM08 – Steering and braking (x1)
- SEM09 – Driver environment, ergonomics, and interfaces (x2)

HELICOPTER EMERGENCY RESCUE (HER25)

Helicopter Emergency Medical Services (HEMS) are crucial for providing rapid medical response, especially in situations where ground transport is too slow or inaccessible. HEMS plays a vital role in saving lives by quickly reaching accident scenes, transporting critically ill patients, and ensuring timely medical interventions. It is particularly essential in rural, mountainous, or congested urban areas where rapid evacuation can make the difference between life and death.

Current HEMS aircraft are existing helicopters retrofitted for medical missions. While highly capable, they have design drawbacks that limit their effectiveness, such as limited cabin space, noise and vibration issues, performance limitations (in hot-and-high conditions or adverse weather), range constraints, operating and maintenance costs, safety concerns and environmental constraints (e.g. noise levels).

The aim of this project is to design the perfect rotorcraft for HEMS operations, providing optimised cabin layout for medical operations, enhance safety features, performance improvements, noise and vibration reduction, cost and maintenance efficiency and all-weather operations.

Academic Advisors: Drs Armanini, Fasel, Ribera-Vicent (coordinator), Seifikar, Shamsuddin and Steiros.

Roles Available:

- HER01 – Project coordination (x2)
- HER02 – Aerodynamics (x5)
- HER03 – Structural Design, Manufacturing and CAD (x4)
- HER04 – Systems, Propulsion and Mission equipment(x5)
- HER05 – Flight Mechanics, Performance & Control (x4)

NEXT GENERATION TRAINER (NGT25)

The aim of this project is the development of a next-generation intermediate/advanced trainer aircraft for the Royal Air Force which can deliver the performance necessary to train aviators to fly next-gen fighters, at significantly lower emission and operating costs.

The resulting aircraft must meet all relevant airworthiness and safety requirements. It must accommodate two flight crew, achieve at least transonic speeds and be highly maneuverable. Provisions for an armed, multirole light attack variant, should be made.

Academic Advisors: Drs Eastham, Levis (coordinator), Quino Quispe and Rigas, Profs Robinson and Vincent

Roles Available:

- NGT01 – Project coordination (x2)
- NGT02 – Aerodynamics (x4)
- NGT03 – Structural Design, Manufacturing & CAD (x4)
- NGT04 – Systems & Propulsion (x4)
- NGT05 – Flight Mechanics, Performance & Control (x4)
- NGT06 – Flight Simulation & Testing (x2)

3.2 How to Make Your Selections – The Rules

Student selections will be made via the dedicated Sharepoint form no later than **Monday 28th April 2025**.

<https://imperiallondon.sharepoint.com/sites/foe/aero/student-portal/Lists/GDP%20Role%20selection/AllItems.aspx>

Indicate your choices by entering the *codes* for the roles you are interested in, in order of preference. Please make sure that you make your choice of roles carefully as mistakes cannot be rectified after the allocation has been made.

Students on the H401 Aeronautical Engineering programme may submit up to 20 choices. To ensure that all students receive one of their chosen projects, you must select **at least two roles from each of the five projects**.

Students on the H415 Spacecraft Engineering programme may only choose from roles in the space project (NEMOX) indicating what their order of preference of all 5 available roles is.

The allocation will be broadly based on students' past performance in the course, while trying to maximise student satisfaction and making sure all roles are filled. The choices of students who fail to follow the rules above will be ignored at first pass, and will be allocated whatever role remains.

4. During the Project

The projects will start on Monday 12th May 2025. Each team will hold a kick-off meeting, to be scheduled in discussion with the supervising academics, where students will be given more information about the project scope and aims. Students should also take this opportunity to meet with their individual supervisors to clarify their individual objectives within the project and set goals for the following week.

Each group will be assigned a room in which they can work and hold meetings. Should there be a shortage of available rooms, groups may have to share a single room and should therefore coordinate their activity timetables appropriately.

4.1 Team Organization

Although smaller sub-groups or individual students may be responsible for only one disciplinary aspect of the design, the entire group's success in this multidisciplinary design project relies on effective communication and collaboration of all group members.

In recognition of this, each group will typically have some students assigned or elected to fill a project management role. These students will act as coordinators to facilitate the running of the project. They will also act as mediators to solve design conflicts that may arise between individual member's design choices, but if an external decision is needed, it will need to be taken by the whole team through an appropriate voting mechanism.

To ensure that all students are kept aware of the group's progress, and how that might have an impact on their areas of responsibility, it is recommended that a short update meeting is held daily, where progress can be presented, design decisions discussed and new goals can be set.

4.2 Weekly Progress Meetings

Each team is expected to hold a weekly progress meeting where all group members and supervising academics will be in attendance. These meetings will be called by the project managers, who will organize the agenda according to the issues that need discussion. A schedule should be available to the academic advisors at the start. Times and dates are to be set in discussion with the supervising academics.

These meetings will offer both students and academics the opportunity to review the past week's progress and discuss any significant problems with their colleagues and staff, in order to reach generally acceptable decisions.

Students will prepare presentations for the weekly group meeting and will be given 10 minutes to present, followed by a 5-minute Q&A. Note that, not all students need to present each week, however each group member must have presented at least twice by the end of the project.

The content of these presentations will in part inform each student's Carrying Out mark, but will also offer an opportunity for students to receive feedback on their work and presentation skills. In addition to the feedback received during these presentations, students should aim to meet with their individual supervisors once a week, following the progress meetings or at a mutually convenient time.

4.3 Budget/Purchasing Guidelines

Each group has been given a £350 budget for any materials, printing or other resources they require. Please note that:

- The budget is meant for the purchase of consumables or other resources required for the successful completion of your project. It may not be used for food and drink.
- Each group should have no more than two people responsible for requesting the Purchase Orders or buying and submitting Expense Claim forms.
- The use of personal Credit Card may be allowed for purchase up to £100. Written approval (via email) from your supervisor is required before purchasing.
- All Expense Claim MUST BE submitted within 3 months of cost incurred.

- Before submitting an expense claim ensure all receipts are itemised; invoice or receipts are attached (credit card slips or bank statement alone will not be sufficient).
- If the group would like to buy items for the amount above £ 100 with suppliers registered on the College system, please send the Purchase Order request email to aero.finance@imperial.ac.uk.
- In exceptional situation, if the Group (person) need to buy anything for the amount above £100 and the supplier is not registered with the College the request for Virtual Credit Card can be sent to aero.finance@imperial.ac.uk. Please check the guidance.
- Account code – quote Project and Group (if applicable).

In addition to the above budget for consumables, each team will be allocated a budget of £15 per person to be used towards a single team building meal.

The linked documents will be helpful when [requesting a virtual credit card](#), [requesting a purchase order](#) to be raised, and filling [expense claims](#).

If you have any questions regarding purchasing items for your project, please contact aero.finance@imperial.ac.uk

5. Assessment

5.1 Project Outcomes / Deliverables

Students are expected to deliver the following pieces of work by the end of the project to report the project outcome and their contributions towards it.

- **Design specification document:** A single collaborative document which details the overall results of the project and the contributions of each subgroup towards achieving them. It can be considered as a summary of the entire project to which each member contributes and should be no longer than 5 pages in total.
- **Poster:** An A1 landscape poster or equivalent video, presenting a top level view of the projects result, detailing in an approachable way the most important design features and characteristics.
- **Individual report:** A technical report by each member of the group detailing their contribution to the project.
- **Group presentation:** A group technical presentation, presenting, detailing and discussing the resulting design.

Some projects may require additional deliverables such as a live demonstration, or demonstration videos from particular sub-groups.

5.2 Overall Project Weighting

Carrying Out

– diligence, innovation, contribution to group (*)	30%
– peer assessment, poster	5%
Individual Report	50%
Group Oral Presentation	15%
Total	100%

(*) as demonstrated throughout the project from interaction with supervisors and during weekly meetings

5.3 Project Presentation

The project presentations for all groups will be held on **Thursday June 19th 2025**. Each group's technical presentation should be 1.5 hours long and will be followed by a 30-minute Q&A session.

The presentation should provide an overview of the students' proposed solution to the project brief. In particular, it should detail the students' approach to the problem, identify major design decisions and the reasoning behind them, present and discuss major results and assess the efficacy of the proposed solution. The most appropriate format, content and breakdown of the presentation are to be determined by the students.

Additionally, each group will have to prepare a 5-minute pitch of their concept. Departmental staff, other students and members of the Departments industrial advisory board will be in attendance. The presentation should present an overview of the work done by each GDP group but will not be assessed.

Students may use the Audio-Visual facilities in any of the lecture rooms at any time when free to practice your presentation (room timetables on doors).

5.4 Presentation Marking

Presentations are marked on the following criteria -

Quality of viewfoils: 20%
Clarity of presentation: 20%
Explanation of Problems: 20%
Technical Challenge & Solution: 20%
Demonstrated level of Group Work: 20%

5.5 Project Report Guidelines

Each student is required to present the results of their research project in a report, the deadline for which is **Monday June 16th 2025**.

The report should detail your contribution to the project and address and justify design choices made throughout the project. A typical report should include an abstract, brief introduction and theory, results, discussion and conclusion sections as necessary to best present your work. A standard template is not provided; instead a common template for the report should be chosen and used by the entire group.

The report should be no longer than 12 pages. The cover page, Tables of Contents and Appendices are not included in such limit. Markers are instructed to not read beyond the page limit and ignore excess pages when marking.

Advice on report writing can be found in the Student Handbook, but students are advised to consult projects from previous years to see the standard expected. In particular, look at reports which in the past have won prizes. Students can view past GDP reports on Sharepoint.

There are penalties for late submission of project work on exactly the same basis as those for coursework (see "Coursework Submission Penalty Scheme" section in your student handbook).

5.6 Carrying Out Marking

The student is also assessed on their conduct (Carrying Out) of the project. This mark is produced at the end of the project. A student who communicates well with the supervisor and group, attends meeting, takes initiative to think about problems before taking them to the supervisor, reads around the topic area, is careful and efficient with calculations, programming and/or experimental procedure, and (most importantly) takes an interest in and makes an important contribution to the project, can expect to get an very good Carrying Out mark.

The peer-assessment component will be based on peer-reviews to be completed on a weekly basis by each project participant

5.7 Mark Schemes

To ensure uniformity of assessment a standard system of grading is to be used. This is based on the criteria given in the mark schemes attached to the end of this document.

6. Project Submissions

Submission Information for Project Reports and Posters

Final Report – Monday 16th June 2025

- ONE pdf file of your individual report to be submitted via your report submission box on Blackboard.

Executive Summary – Monday 16th June 2025

- ONE pdf file of the group's design specification document to be submitted via the Executive Summary submission box on Blackboard (no hard copies to be submitted with your reports).

Group Poster – Thursday 19th June 2025

- ONE pdf of an A1-size poster to be submitted via the Poster submission box on Blackboard. You may wish to print a copy for your group presentation and/or pitch.

Carry-Out Mark-Scheme

Criteria	0-39% (F, D-) Poor	40-49% (D, D+) Satisfactory	50-59% (C-, C, C+) Good	60-69% (B-, B, B+) Very Good	70-89% (A-, A, A+) Excellent	90-100% (A*) Outstanding
<i>Contribution to Group</i> [20%]	Student demonstrated little to no ability to work as part of a group. Participation to group meetings was extremely limited	Student demonstrated a very limited ability to work within a group however did participate in some group meetings.	Student demonstrated a limited ability to work within a group. Nevertheless they attended most meetings and made an effort to communicate with peers. .	Student demonstrated an ability to effectively work within a group. They took part in meetings, contributed to the success of the project but largely concentrated on own work.	Student demonstrated an excellent ability to work with others. They took an active role in the group's work, contributing to its overall success. They actively contributed to group discussions and helped enrich the work of peers.	Very little could be done to better it in any way
<i>Technical Skills</i> [20%]	Student demonstrated little to no ability in developing and utilizing any computational, analytical and/or experimental tools necessary for their project.	Student demonstrated a limited ability to develop and use the computational, analytical and/or experimental tools provided for their project.	Student demonstrated an ability to develop and use the computational, analytical and/or experimental tools for necessary for their project.	Student demonstrated the ability to identify appropriate computational, analytical and/or experimental tools necessary for their project, effectively developing and utilizing them with little support.	Student demonstrated an excellent ability to identify the most appropriate computational, analytical and/or experimental tools necessary for their project, effectively developing and utilizing them. Minimal to no support was required.	
<i>Originality</i> [10%]	No originality in the approach selected and employed. Work presented is of a poor standard.	No originality in the approach employed however reasonable, yet simplistic methods were employed and work is of satisfactory standard.	While not original in its approach, the methods used were mostly appropriate and the work presented was of a good standard.	The methods employed were selected with appropriate considerations and were entirely appropriate to the problem at hand.	A good mix of established methods with some Innovative approaches were employed. Choice of methods was well reasoned and entirely appropriate	
<i>Diligence</i> [30%]	No significant student effort was evident throughout the duration of the project.	Some student effort was evident throughout the duration of the project, however falling well short of the expected level.	Student worked diligently throughout the project with effort however failing to meet the expected level.	Student worked diligently throughout the project with effort up to the expected level.	Student worked diligently throughout the project with effort well beyond the expected level.	
<i>Reporting</i> [20%]	Student provided few or no updates on their progress. Limited information was provided and generally not in an appropriate format. Poor communication skills overall.	Student provided some updates on their progress, in a less-than regular basis, important information was often missing and significant probing by the supervisor was necessary. Limited communication skills evident.	Student provided good, regular updates on their progress, however some important information was often missing and significant probing by the supervisor was necessary.	Student provided very good, regular updates on their progress, however some probing by the supervisor was necessary. Very good communication skills overall.	Student provided excellent, regular updates on their progress in an appropriate format. Little to no probing by the supervisor was necessary. Excellent communication skills overall.	

Report Mark-Scheme						
Criteria	0-39% (F, D-) Poor	40-49% (D, D+) Satisfactory	50-59% (C-, C, C+) Good	60-69% (B-, B, B+) Very Good	70-89% (A-, A, A+) Excellent	90-100% (A*) Outstanding
<i>Structure of Report</i> [12.5%]	Document is poorly presented throughout indicating limited effort	Significant issues are present with the presentation of the work. Captions and presentation of figures, tables and equations. Numerous grammatical/spelling errors	Document is correctly formatted; Abstract is limited and some issues with captioning and presentation of figures, tables and equations. Some grammatical/ spelling errors.	Document is correctly formatted; Minor issues with presentation of sections, captioning and presentation of figures, tables and equations.	Document is very well presented and complete. A pleasure to read. All figures, tables and equations correctly labelled and referred.	Very little could be done to better it in any way
<i>Balance of Sections</i> [12.5%]	Key sections missing, indicating a limited understanding of the scope of the project and how to best communicate the work done.	Report is split into typical sections, however they are not entirely appropriate to presenting the work that has been completed	The reports is split into an appropriate set of sections, the relative length and detail of which is not appropriate to the scope of the project.	The reports is split into an appropriate set of sections, the relative length and detail of which is mostly appropriate to the scope of the project.	The reports is split into an appropriate set of sections, the relative length and detail of which is appropriate to the scope of the project, clearly communicating the work done.	
<i>Clarity of Argument</i> [12.5%]	Demonstrates incomplete or incorrect interpretation of trends indicating lack of understanding of results	Demonstrated limited thoughts and arguments for choices made; limited explanations of findings and ideas from results and literature; limited argument and discussion of the work.	Demonstrated a reasonable level of understanding of the phenomena investigated; some key findings and choices made are related to ideas from literature; sustains some modest arguments; present some discussion of the work as a whole.	Demonstrated clear thoughts and arguments; explained findings and choices made, supported by concepts and findings from literature; provides clear and sound arguments that are well defended; presents a well round discussion of the work as a whole.	Demonstrated an independence of thought and arguments, indicating wide exploration; explains rationale for choices made clearly and provides lucid, cogent and sophisticated arguments that are well substantiated and defended.	
<i>Validity of Results</i> [12.5%]	Presents poorly constructed set of results in inappropriate graphical, figurative or tabulated forms. The results are however mostly wrong and invalid.	Presents limited results in reasonable graphical, figurative or tabulated forms. Results are imprecise and to be questioned.	Presents most relevant results in a mostly appropriate graphical, figurative or tabulated form. Results contain errors and are sometimes elusive.	Presents all results in the most appropriate form. Results are mostly correct but minor issues identified.	Presents all results in in the most appropriate form allowing easy comparison and comprehension of trends. Results are accurate throughout.	
<i>Justification of Conclusions</i> [12.5%]	Demonstrates poor conclusions; no evidence or attempt to answer the project objectives.	Demonstrated limited conclusions; limited evidence on attempt to use findings or literature to support them.	Demonstrated modest conclusions; presenting some answers to the project objectives. Findings and literature provides limited support for the conclusions presented	Demonstrated clear conclusions; presents answers to the project objectives, mostly evidenced by their findings and the literature.	Demonstrates concise and reasonable conclusions; presents coherent answers to the project objectives, grounded in their findings and literature demonstrating a thorough understanding of the topic.	
<i>Details of Presentation</i> [12.5%]	Little to no information presented in support of the arguments made.	Significant gaps in the results presented greatly limiting the reader's ability to understand the author's work.	Some information missing and/or not presented in an appropriate format, limiting the reader's ability to understand some aspects of the authors work.	All key information is presented but some details missing, limiting the reader's ability to understand the entirety of work done.	All information necessary to support the arguments made is presented and is in the most appropriate format.	
<i>References & Bibliography</i> [12.5%]	A poorly organised, incomplete reference list not following appropriate style.	Limited, yet structured list of references. An appropriate style is not consistently followed	A modest, organised and structured reference list. Some small errors encountered.	An appropriate and clearly organised and structured reference list. Minor issues in style used.	A well developed bibliography entirely appropriate to the topic. List of references is organised and structured with sufficient information given to easily retrieve source given	
<i>Achievement of Objectives</i> [12.5%]	Little or no serious work, failure to address task; most work packages not addressed.	Little actual achievement, limited understanding.	Some weaknesses and limited in achievement, but implementation was diligent and major objectives have been addressed.	High standards, which, though failing to match up to total expectations nevertheless contained much actual achievement.	Highly competent work, with achievement up to or beyond the supervisor's expectation.	

